Requirement Analysis Report

E-Commerce Order Management Database System

Project Phase: Week 1 – Understanding the Business Problem

Role: Database Analyst

Document Type: Requirement Analysis Report

1. Introduction

This report documents the requirement analysis carried out for the E-Commerce Order Management Database System. The purpose of this analysis is to study the business operations of an online shopping platform, understand how data flows between its different functions, and translate that understanding into a structured set of requirements that will guide the database design in the subsequent phases of the project (ER modeling, normalization, SQL implementation, and reporting).

2. Business Scenario Overview

A rapidly growing e-commerce company needs a centralized database system to run its day-to-day operations. The platform must support customer registration, product and category management, supplier coordination, inventory tracking, order processing, payment management, shipment tracking, and customer reviews. The core business goal is to replace fragmented or manual record-keeping with a single, accurate, and consistent data source that reduces redundancy, improves operational efficiency, and enables meaningful business reporting for decision-making.

3. Purpose of Each Core Entity

The system is built around ten mandatory core entities. Each entity represents a distinct real-world object or transaction in the business and together they capture the full lifecycle of an order, from customer sign-up to product review.

Entity	Purpose in the System
Customer	Stores the personal and contact details of registered users (Customer ID, Name, Email, Mobile Number, Address, Registration Date). It uniquely identifies who places orders and writes reviews.
Category	Groups products into logical classifications (Category ID, Category Name, Description) to support browsing, filtering, and catalog organization.
Product	Holds the sellable catalog items (Product ID, Product Name, Price, Stock Quantity, Category ID). It links to Category and is the item referenced by orders and reviews.
Supplier	Records vendors who supply products to the business (Supplier ID, Supplier Name, Contact Information), supporting procurement and stock replenishment.

Entity	Purpose in the System
Order	Represents the header of a customer transaction (Order ID, Customer ID, Order Date, Total Amount, Order Status), tying together everything purchased in a single checkout.
Order Details	Captures the individual line items within an order (Order Detail ID, Order ID, Product ID, Quantity, Unit Price), allowing one order to contain many products.
Payment	Tracks how and when an order was paid for (Payment ID, Order ID, Payment Method, Payment Date, Payment Status), supporting reconciliation and financial reporting.
Shipment	Manages delivery logistics for an order (Shipment ID, Order ID, Delivery Address, Shipment Date, Delivery Status), enabling order fulfillment tracking.
Review	Stores customer feedback on purchased products (Review ID, Customer ID, Product ID, Rating, Comments), supporting product quality insight and trust-building.

4. Major Business Processes

Analysis of the scenario identifies eight major business processes that the database must support end to end:

- **Customer Registration and Profile Management** – capturing new customer sign-ups and maintaining contact details.
- **Category and Product Catalog Management** – organizing products into categories and maintaining pricing and descriptions.
- **Supplier and Inventory Management** – onboarding suppliers and tracking stock quantity to prevent overselling or stockouts.
- **Order Placement and Processing** – creating an order, recording ordered products, quantities and unit prices, and updating order status.
- **Payment Processing** – recording the payment method, date, and status against each order.
- **Shipment and Delivery Tracking** – recording the delivery address, shipment date, and delivery status of each order.
- **Customer Review and Feedback Management** – letting customers rate and comment on products they have purchased.
- **Business Reporting and Analytics** – generating summaries such as sales trends, best-selling products, and order status reports from the accumulated data.

5. Stakeholders and Their Roles

Stakeholder	Role
Customer	End user who registers, browses products, places orders, makes payments, and submits reviews.

Stakeholder	Role
Store/Business Administrator	Manages product catalog, categories, pricing, and monitors overall order and inventory activity.
Inventory / Warehouse Staff	Maintains stock quantity accuracy and coordinates restocking with suppliers.
Supplier / Vendor	Provides products to the business and is a source of restocking information.
Order Processing / Sales Team	Handles order confirmation, status updates, and resolves order-related issues.
Finance / Payment Team	Monitors payment status, reconciles transactions, and manages refunds or failed payments.
Logistics / Delivery Team	Manages shipment creation, delivery address verification, and delivery status updates.
Database Administrator / Developer	Designs, implements, and maintains the underlying database and ensures data integrity and security.
Business Owner / Management	Uses reports generated from the system to make strategic and operational decisions.

6. Key Observations from the Analysis

- Every order is tied to exactly one customer, but a customer can place many orders over time — a one-to-many relationship.
- An order can contain multiple products, and a product can appear in many orders, which the Order Details entity resolves as an associative table.
- Payment and Shipment are both dependent on Order, meaning no payment or shipment record should exist without a valid order.
- Product stock quantity must be kept consistent with order activity to avoid overselling.
- Reviews are linked to both Customer and Product, allowing feedback to be traced back to a specific purchaser and item.
- Category and Supplier act as supporting/reference data that keep the Product catalog organized and traceable to its source.

7. Conclusion

The analysis confirms that the ten mandatory entities are sufficient to model the full order lifecycle of an e-commerce platform, from registration through to post-purchase review. This understanding forms the basis for the Business Requirement Document and the Software Requirement Specification that follow, and will directly inform the ER diagram and normalization work in later weeks.